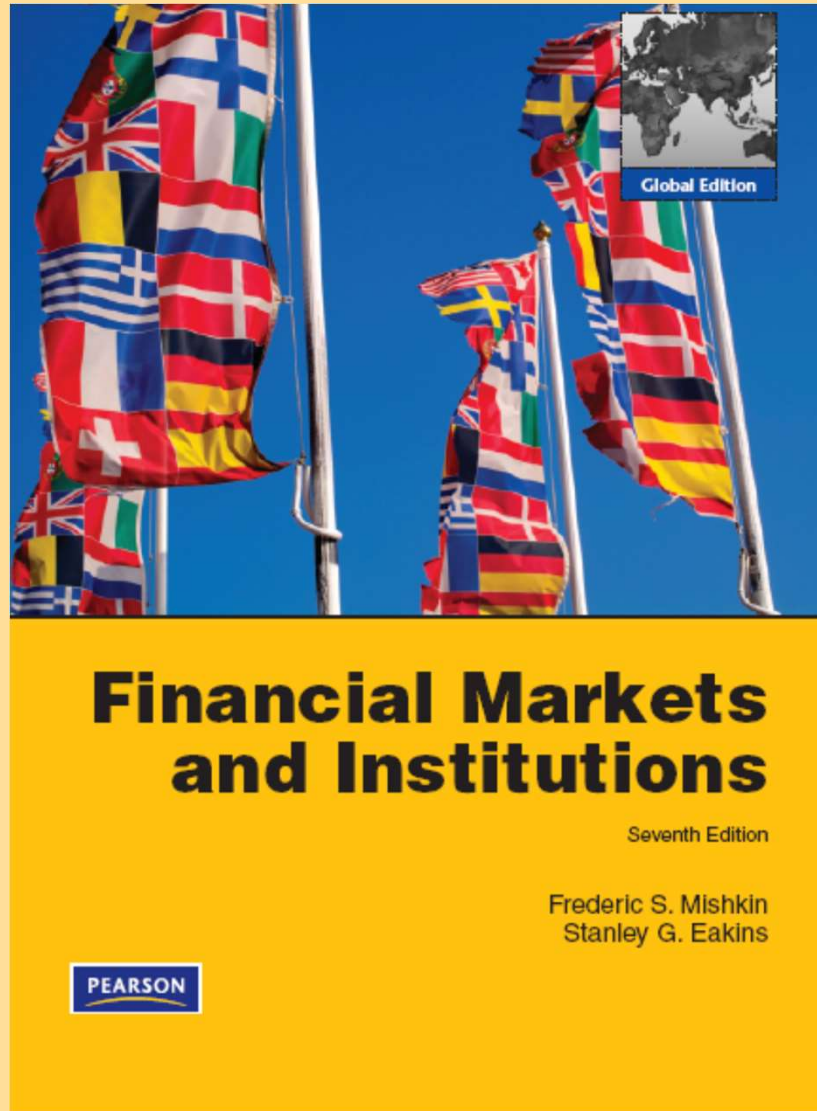


LECTURE 1 (CHAPTER 2)

Overview of the Financial System





Chapter Preview

- “Big picture” question:

WHY DO WE NEED A FINANCIAL SECTOR?

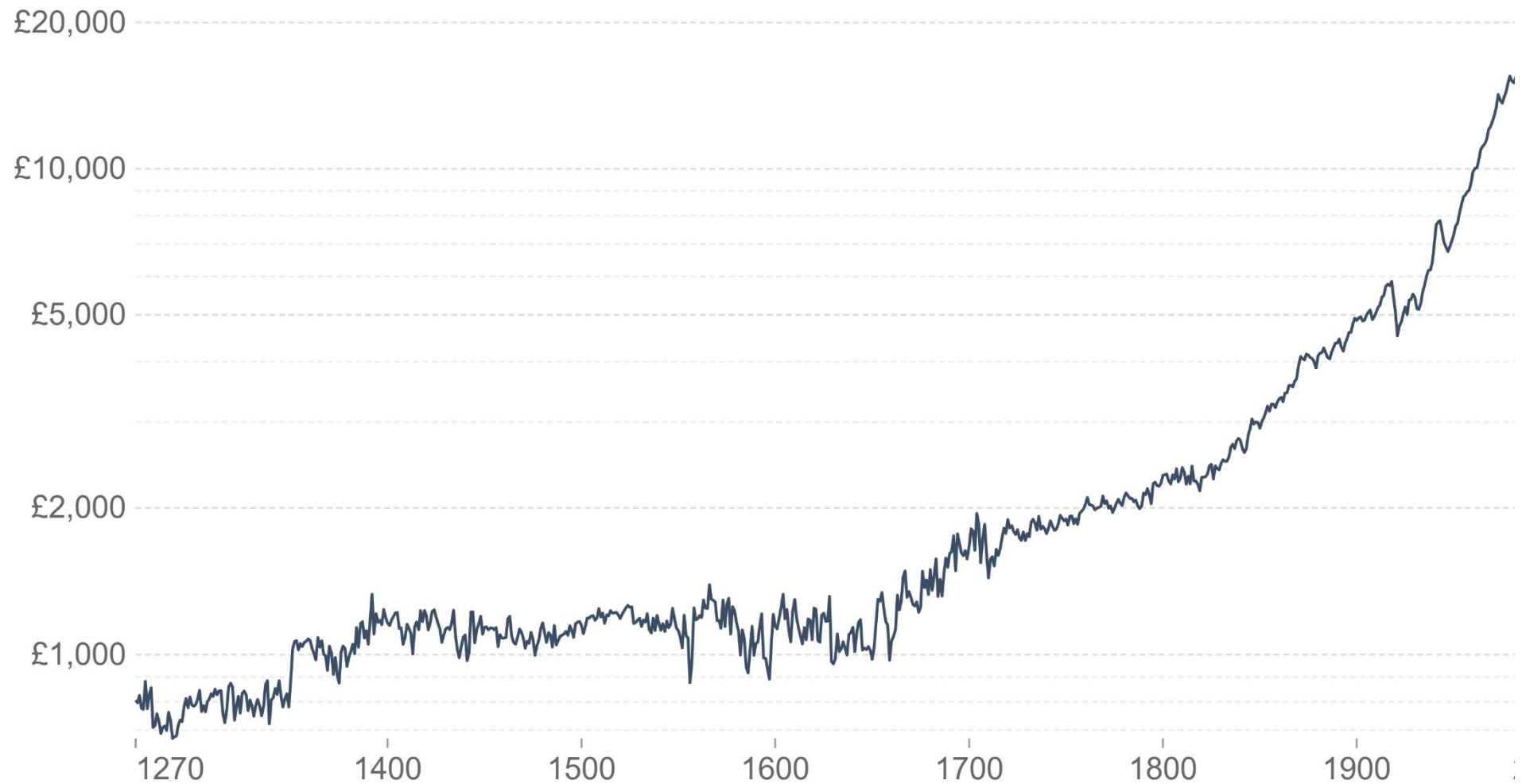
- An even more fundamental question:

■ IS FINANCE GOOD FOR THE ECONOMY?

financial markets transfers money to those who invest better the money NOW

GDP per capita in England

Adjusted for inflation and measured in British Pounds in 2013 prices



Source: Broadberry, Campbell, Klein, Overton, and van Leeuwen (2015) via Bank of England (2020)

Note: Data refers to England until 1700 and the UK from then onwards.

OurWorldInData.org/economic-growth • CC BY



Finance and Economic Growth

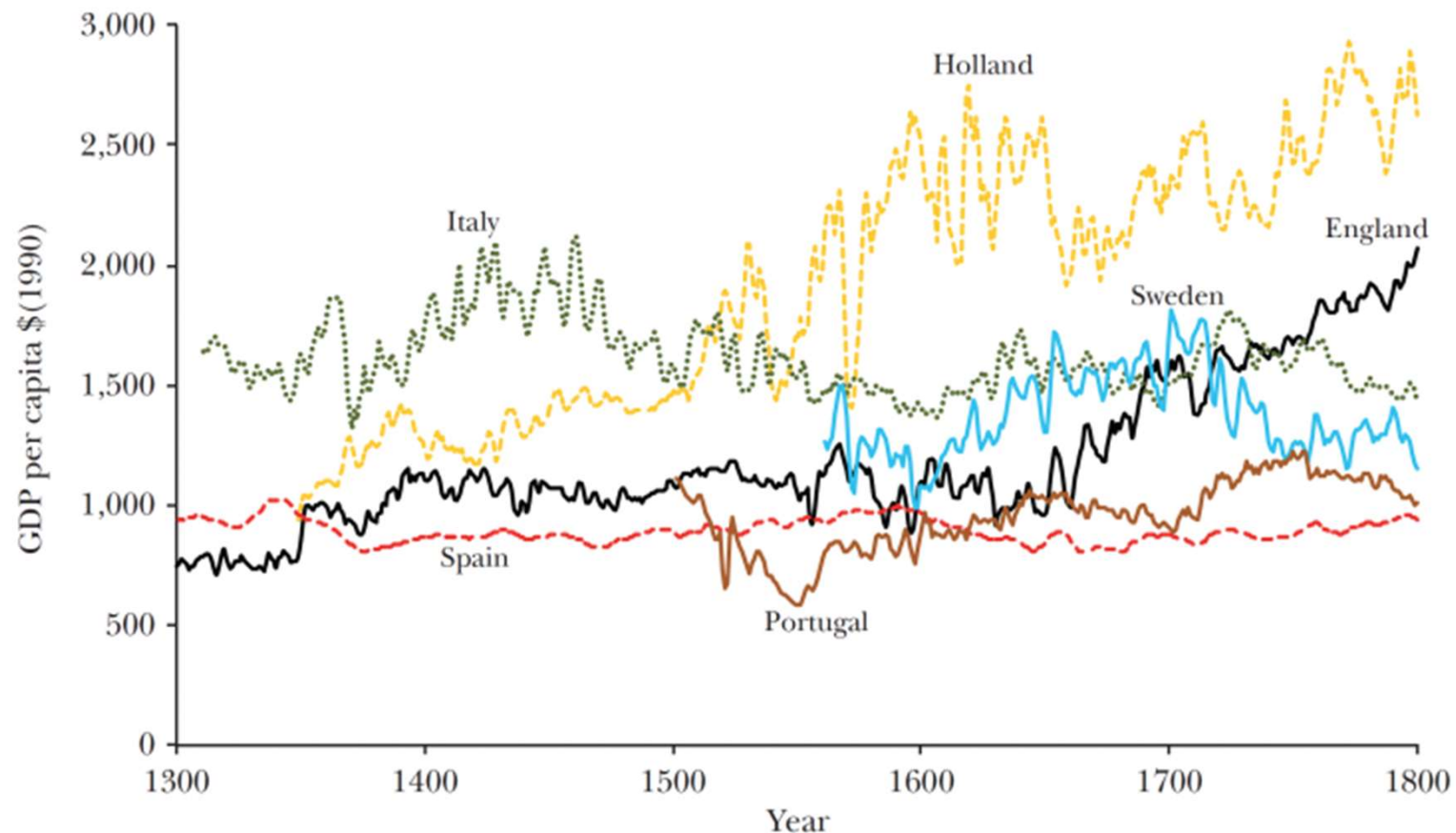
- Throughout most of human history, no (per capita) economic growth
 - Zero-sum game: more people meant lower incomes
- Not because of the lack of **technological innovations**: windmill, crop-rotations, even new crops from the Americas...
 - These pushed incomes up but only temporarily
- Sustained growth per capita only really picked up after 1650. Why?



Growth in North-West EU

GDP per Capita in Selected European Economies, 1300–1800

(three-year average; Spain eleven-year average)





The role of finance

- **Finance played a major role:** without financing, difficult to invest in infrastructure, etc
- Drain of natural and human resources from the colonies helped. But not the only factor: otherwise Spain would have been the richest country of all
- Finance **is a promise to future payments** (credit comes from Latin *to trust*)
- Trade by Dutch and British merchants required financing (and insurance)
- But the system can be sustained only with a **belief in growth**: otherwise credit (with interest) is not possible



The Benefit of Financial Markets: An Example

- Suppose you want to start a business manufacturing a household cleaning robot, but you have no funds.
- Walter has money he wishes to invest for his retirement.
- If the two of you could get together, perhaps both of your needs can be met. But how does that happen?



Importance of Financial Markets

- Financial markets help to efficiently allocate capital by transferring funds from people who lack investment opportunities to people who have them.
- Financial markets also improve the welfare of consumers, allowing them to better time their consumption.



Key Concept: Asset

Real Asset:

concrete entity that last over time

Entity producing flow of **goods or services**

Examples:

Land

Plants, machinery

People

New invention

Goodwill, reputation, brand (not tangible)

finance matches real asset with financial asset

- Borrowers **sell or issue** financial assets to lenders and use the proceeds **to buy real assets** (an investment)

Financial Asset:

human convention that can be worth nothing if the issuer does not recognize them anymore

Contract giving its owner a **claim to payments**

Examples:

Currency

Bonds, stocks

Bank deposits, loans

Insurance contracts

Derivatives (futures, options...)



Segments of Financial Markets

1. Direct Finance

- Borrowers raise funds directly by selling financial assets to lenders. These assets are claims to the borrower's future income from human and nonhuman real assets

2. Indirect Finance

- Borrowers borrow indirectly from lenders via financial intermediaries, which issue financial instruments which are claims on the borrower's future income from human or nonhuman real assets

why do we need a bank?

Bank have expertise, they better know which contract to issue.
Diversified

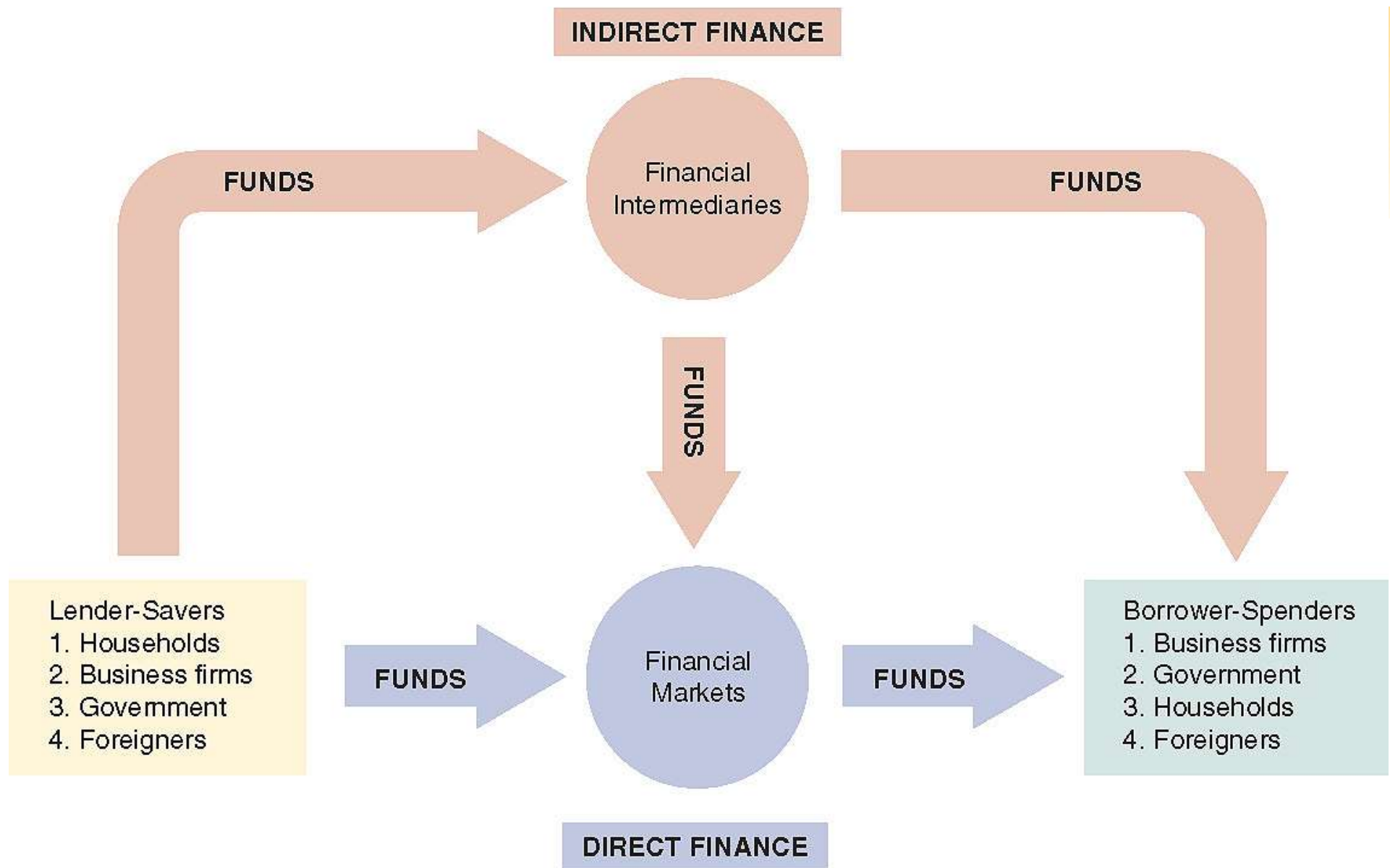


FIGURE 2.1 Flows of Funds Through the Financial System

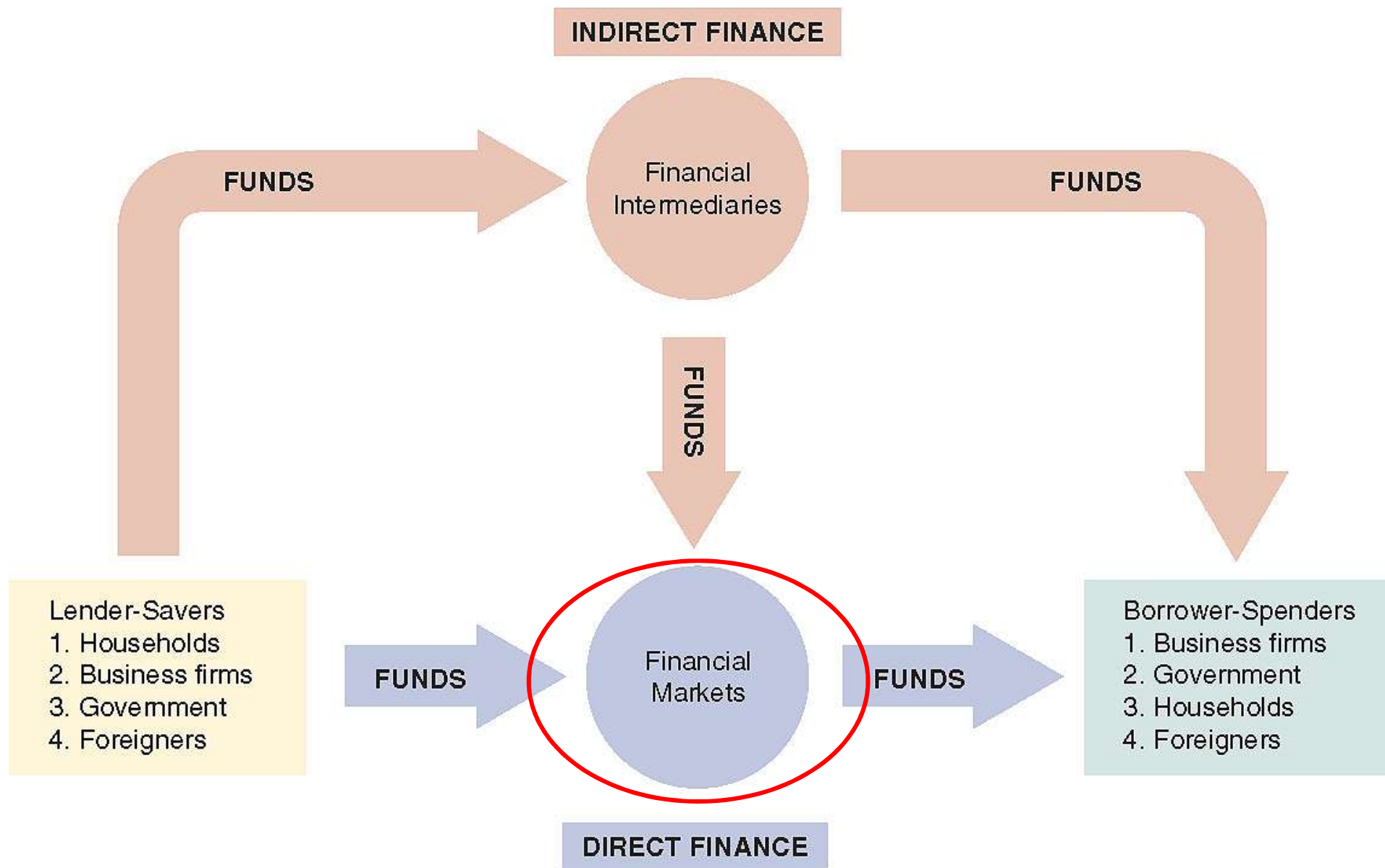


FIGURE 2.1 Flows of Funds Through the Financial System



Classifications of Financial Markets (I)

1. Debt Markets

- Short-Term (maturity < 1 year)
- Long-Term (maturity > 10 year)
- Intermediate term (maturity in-between)
- Represented \$52.4 trillion at the end of 2009.

2. Equity Markets

- Pay dividends, in theory forever
- Represents an ownership claim in the firm
- Total value of all U.S. equity was \$20.5 trillion at the end of 2009.



Classifications of Financial Markets (II)

1. Money Market: Short-Term
(maturity < 1 year)
2. Capital Market: Long-Term
(maturity > 1 year) plus equities



Classifications of Financial Markets (III)

1. Primary Market

- New security issues sold to initial buyers
- Typically involves an investment bank who underwrites the offering

2. Secondary Market

- Securities previously issued are bought and sold
- Examples include the NYSE and Nasdaq



Secondary Markets (I)

Even though secondary markets do not directly channel money to productive investment, they are important for two reasons:

- Provide liquidity, making it easy to buy and sell the securities of the companies (allow self financing through asset accumulation)
- Establish a price for the securities in primary mkts



Function of Financial Intermediaries: Indirect Finance

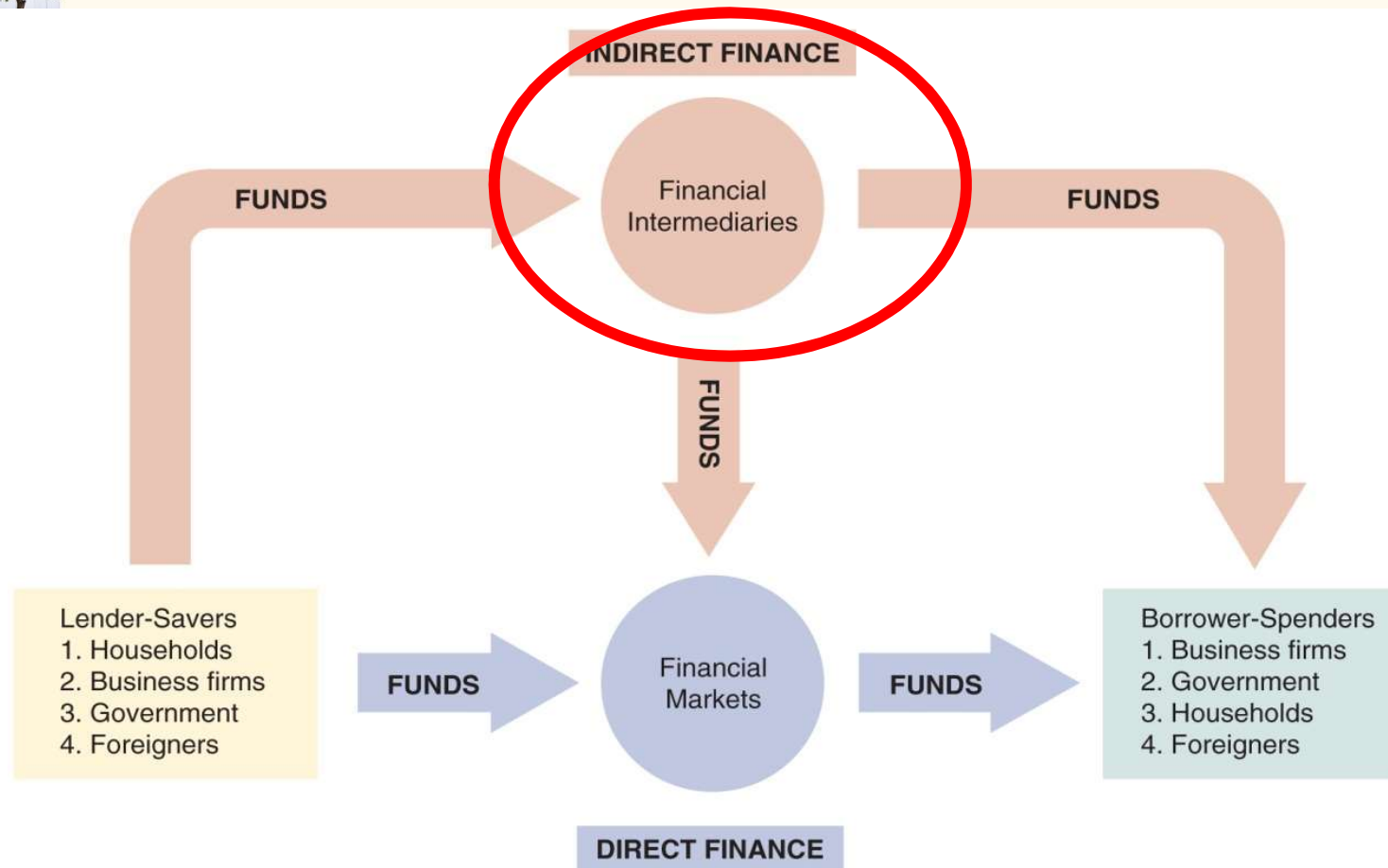


FIGURE 2.1 Flows of Funds Through the Financial System



Function of Financial Intermediaries: Indirect Finance

Instead of savers lending directly to borrowers, a financial intermediary (such as a bank):

- the intermediary obtains funds from savers
- the intermediary then makes loans/investments with borrowers

More important source of finance than securities markets (such as stocks or bonds)



Function of Financial Intermediaries: Indirect Finance

Why do we need financial intermediaries?

Why do they have a comparative advantage at lending (or borrowing)?

- Important role of transactions costs, risk sharing, and asymmetric information



Asymmetric Information: Adverse Selection and Moral Hazard

Financial intermediaries reduce adverse selection and moral hazard problems, enabling them to make profits. How they do this is covered in many of the chapters to come.



Types of Intermediaries

Type of Intermediary	Value of Assets (\$ billions, end of year)			
	1990	2000	2010	2015
Depository institutions (banks)				
Commercial banks, savings and loans, and mutual savings banks	4,779	7,687	15,580	17,368
Credit unions	215	441	912	1,146
Contractual savings institutions				
Life insurance companies	1,367	3,136	5,176	6,350
Fire and casualty insurance companies	533	862	1,242	1,591
Pension funds (private)	1,629	4,355	4,527	8,517
State and local government retirement funds	737	2,293	2,661	5,641
Investment intermediaries				
Finance companies	610	1,140	1,439	1,474
Mutual funds	654	4,435	7,935	12,843
Money market mutual funds	498	1,812	2,755	2,716